A Multivariate Analysis of the Physicochemical Characteristics of Portuguese Vinho Verde Wines

1. Introduction

This study investigates the physicochemical properties of Portuguese Vinho Verde wines to understand the underlying structural dimensions of wine chemistry and differentiate between red and white variants. Understanding these characteristics is essential for quality control and sensory evaluation. The purpose of this study is to apply multivariate statistical techniques to explore group differences, reduce dimensionality, and identify latent factors driving wine composition.

2. Dataset

The analysis utilizes the Wine Quality Dataset from the UCI Machine Learning Repository. The combined dataset consists of 6,497 total observations, comprising 1,599 red wine samples and 4,898 white wine samples. It includes 11 continuous physicochemical variables, a categorical variable for wine color, and an integer-based sensory quality score.

3. Research Questions and Hypotheses

Research Question 1 (MANOVA): Do red and white wines differ significantly in their overall physicochemical characteristics when several characteristics are considered simultaneously?

- **H0:** The mean vector of the selected physicochemical variables is the same for red and white wines.
- **H1:** The mean vectors differ between red and white wines.

Research Question 2 (PCA): Can the 11 physicochemical measurements of wine be summarized using a smaller number of principal components while retaining a substantial proportion of the total variability?

Research Question 3 (EFA): What underlying latent dimensions or factors explain the relationships among the physicochemical characteristics of wine?

4. Methodology

Data preparation, visualization, and statistical modeling were conducted using Python. Exploratory Data Analysis (EDA) was used to assess distributions, correlations, and outliers. A Multivariate Analysis of Variance (MANOVA) was conducted to test for combined mean differences between wine colors. Principal Component Analysis (PCA) was applied to standardized data to reduce dimensionality, and Exploratory Factor Analysis (EFA) with a Promax rotation was used to identify latent chemical constructs.

5. Results

5.1. Exploratory Data Analysis

- **Variability & Correlation:** An examination of the descriptive statistics revealed that total sulfur dioxide (SD = 56.52) and free sulfur dioxide (SD = 17.75) exhibited the highest variability among the continuous variables. The correlation matrix highlighted several strong relationships, most notably a strong positive correlation between total and free sulfur dioxide ($r = 0.72$) and a strong negative correlation between density and alcohol ($r = -0.69$).
- **Differences:** Visual inspections via boxplots indicated notable disparities between the wine colors. Red wines displayed substantially higher levels of volatile acidity (mean = 0.53) compared to white wines (mean = 0.28), while white wines generally contained higher free sulfur dioxide and residual sugar levels.

5.2. MANOVA

- **Assumptions:** Prior to analysis, data were evaluated for multivariate normality and homogeneity of covariance matrices. While some minor deviations in normality were present, MANOVA is robust to such violations given the large sample size ($N = 6,497$).
- **Findings:** A MANOVA was conducted utilizing fixed acidity, volatile acidity, citric acid, and residual sugar as dependent variables to investigate differences between red and white wines. The results demonstrated a highly statistically significant difference between wine colors across the combined variables (Wilks' Lambda = 0.4026, $F = 2408.72$, $p < 0.001$).
- **Interpretation:** This finding indicates that red and white Vinho Verde wines possess fundamentally distinct chemical profiles. Rather than varying on just a single metric, the two wines represent distinct multivariate populations largely driven by differing acidity and fermentation structures.

5.3. Principal Component Analysis (PCA)

- **Component Retention:** Based on an examination of the scree plot and the cumulative variance explained, four principal components were retained. Together, these four components account for 73.19% of the cumulative variance within the original 11 physicochemical variables.
- **Interpretation:** The varimax-rotated component loadings suggest distinct chemical profiles:
    - **PC1 (Sulfur/Preservative Dimension):** Heavily loaded by total sulfur dioxide and free sulfur dioxide.
    - **PC2 (Density/Alcohol Profile):** Heavily loaded by density (positive) and alcohol (negative).
    - **PC3 (Volatile Acidity Profile):** Heavily loaded by volatile acidity and pH.
    - **PC4 (Fixed Acidity Profile):** Heavily loaded by fixed acidity and citric acid.

5.4. Exploratory Factor Analysis (EFA)

- **Factorability:** The data were deemed highly suitable for factor analysis based on a significant Bartlett's test of sphericity ($p < 0.001$) and a Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy of 0.74, which exceeds the recommended threshold of 0.60.
- **Interpretation:** Using a Promax (oblique) rotation, three underlying latent factors were extracted:
    - **Factor 1 (Preservation & Sweetness):** Driven by total sulfur dioxide, free sulfur dioxide, and residual sugar.
    - **Factor 2 (Acidity & pH Structure):** Driven by fixed acidity, citric acid, and inverse pH.
    - **Factor 3 (Fermentation/Density):** Driven by alcohol and density.

6. Integrated Discussion

1. MANOVA: Red and white wines are distinguished primarily by their acidity profiles. Follow-up univariate ANOVAs revealed that volatile acidity yielded the largest effect size distinguishing the groups ($F = 4829.32$, $\eta^2 = 0.426$), followed by fixed acidity ($F = 2016.51$, $\eta^2 = 0.237$) and residual sugar ($F = 899.77$, $\eta^2 = 0.122$).

2. PCA Dimensions: The major dimensions of variation summarize the dataset into four orthogonal components representing Preservatives, Density/Alcohol, Volatile Acidity, and Fixed Acidity.

3. EFA Latent Factors: The underlying relationships among the 11 variables are driven by three correlated latent factors: Preservation/Sweetness, Acidity Structure, and Fermentation/Density.

4. PCA vs. EFA: Both PCA and EFA effectively reduced the dataset, with strong alignment in how they grouped sulfur variables and acidity variables. However, PCA mathematically forced four orthogonal, uncorrelated components to strictly maximize variance. In contrast, EFA allowed factors to correlate, combining the fixed and volatile acidity into a broader, more theoretically cohesive "Acidity Structure" construct, which is more representative of actual wine chemistry.

5. Substantive Interpretation: Chemically, Vinho Verde wines are not just a random collection of 11 variables, but rather a structured profile driven by a few key dimensions. These underlying constructs heavily dictate the fundamental physicochemical difference between red and white variants.

6. Limitations: Two limitations of this analysis include the imbalanced sample sizes (1,599 red vs. 4,898 white) and the purely observational nature of the dataset. This prevents causal claims regarding how specific chemical adjustments might directly alter other properties like density or perceived sensory quality.

7. Conclusion

This study successfully integrated MANOVA, PCA, and EFA to evaluate Portuguese Vinho Verde wines. The MANOVA confirmed significant, multivariate physicochemical differences between red and white wines, largely driven by volatile acidity. PCA effectively reduced the 11 continuous variables into a concise set of principal components capturing 73.19% of the variance, while EFA revealed the underlying latent chemical structures such as Preservation, Acidity, and Fermentation. Together, these multivariate techniques provide a comprehensive and accurate structural summary of wine chemistry.

8. References

- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, 47(4), 547-553.
- UCI Machine Learning Repository. (n.d.). Wine Quality Dataset. Retrieved from https://archive.ics.uci.edu/dataset/186/wine+quality